Notes on Fourier Transforms

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1 Introduction and Definition

1.1 The Fourier Transform Pair

The Fourier transform is an integral transform that decomposes a function into its constituent frequencies. For a function $f(x)$, its Fourier transform, denoted by $F(s)$ or $\mathcal{F}\{f(x)\}$, is a complex-valued function of a real variable s, which represents the frequency.

Definition 1.1 (Fourier Transform)**.** The **Fourier transform** of a function $f(x)$ is defined as:

$$F(s) = \mathcal{F}\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} \, \mathrm{d}t \tag{1}$$

Definition 1.2 (Inverse Fourier Transform)**.** Given the Fourier transform $F(s)$, the original function $f(x)$ can be recovered through the **inverse Fourier transform**, denoted by $\mathcal{F}^{-1}\{F(s)\}$:

$$f(x) = \mathcal{F}^{-1}\{F(s)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{isx} \, \mathrm{d}s \tag{2}$$

The pair of equations (1) and (2) is known as the Fourier transform pair. The existence of the transform is guaranteed if $f(x)$ is absolutely integrable, i.e., $\int_{-\infty}^{\infty} |f(x)| \, \mathrm{d}x < \infty$. The variable x often represents time or position, while s represents angular frequency.

1.2 From Fourier Series to Fourier Transform (A Heuristic Approach)

The Fourier transform can be seen as the limit of the Fourier series as the period of the function approaches infinity. This provides a natural bridge for students already familiar with Fourier series.

1. **Start with Fourier Series:** Consider a function $f_L(x)$ which is periodic with period $2L$. Its complex Fourier series is:
$$f_L(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{n\pi x}{L}}$$
where the coefficients c_n are given by:
$$c_n = \frac{1}{2L} \int_{-L}^{L} f_L(t) e^{-i\frac{n\pi t}{L}} \, \mathrm{d}t$$

2. **Introduce Frequency:**** Define the angular frequency $s_n = \frac{n\pi}{L}$. The spacing between consecutive frequencies is $\Delta s = s_{n+1} - s_n = \frac{\pi}{L}$.

3. **Rewrite the Series:** Substitute c_n into the series expression for $f_L(x)$:
$$f_L(x) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2L} \int_{-L}^{L} f_L(t) e^{-is_n t} \, \mathrm{d}t \right) e^{is_n x}$$
Using $\Delta s = \pi/L$, we can write $1/(2L) = \Delta s/(2\pi)$.
$$f_L(x) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \left(\int_{-L}^{L} f_L(t) e^{-is_n t} \, \mathrm{d}t \right) e^{is_n x} \Delta s$$

4. **Take the Limit $L \to \infty$:** Now, let's consider a non-periodic function $f(x)$ as a function with an infinite period. As $L \to \infty$:

- The frequency spacing $\Delta s = \pi/L \to 0$. The discrete frequencies s_n become a continuous variable s.
- The sum $\sum \ldots \Delta s$ becomes a Riemann sum, which converges to an integral $\int \ldots \, \mathrm{d}s$.
- The function $f_L(x)$ becomes our non-periodic function $f(x)$, and the integral limits go from $[-L, L]$ to $(-\infty, \infty)$.

The expression becomes:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(t) e^{-ist} \, \mathrm{d}t \right) e^{isx} \, \mathrm{d}s$$

5. **Identify the Transform Pair:** We can now identify the inner integral as the Fourier transform. To match our symmetric definition, we distribute the $1/(2\pi)$ factor as $1/\sqrt{2\pi}$ for each integral.
   - Let $F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} \, \mathrm{d}t$.
   - Then $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{isx} \, \mathrm{d}s$.

   This shows how the concept of representing a function as a sum of discrete sinusoids (Fourier Series) evolves into representing it as an integral over a continuum of sinusoids (Fourier Transform).

1.3 Interpretation and Notes

Note 1.1. A function $f(x)$ is said to be **self-reciprocal** under the Fourier transform if its transform has the same functional form, i.e., $\mathcal{F}\{f(x)\} = f(s)$. A famous example is the Gaussian function $e^{-x^2/2}$.

Note 1.2 (Spectral Representation)**.** If $\mathcal{F}\{f(x)\} = F(s)$, then the inverse transform formula $f(x) = \mathcal{F}^{-1}\{F(s)\}$ is called the **spectral representation** of $f(x)$.

- $F(s)$ is the **spectral density** or **spectrum** of $f(x)$.
- $|F(s)|$ is the **amplitude spectrum**.
- $|F(s)|^2$ is the **energy spectrum** or **power spectral density**.
- s is the **angular frequency**.

2 Properties of the Fourier Transform

Property 2.1 (Linearity)**.** The Fourier transform is a linear operator. For constants c_1, c_2 and functions $f(x), g(x)$:

$$\mathcal{F}\{c_1 f(x) + c_2 g(x)\} = c_1 \mathcal{F}\{f(x)\} + c_2 \mathcal{F}\{g(x)\}$$

Proof: Follows directly from the linearity of the integral.

Property 2.2 (Frequency Shift / Modulation)**.** If $\mathcal{F}\{f(x)\} = F(s)$ and a is a real constant:

$$\mathcal{F}\{e^{iax} f(x)\} = F(s - a)$$

Proof:

$$\mathcal{F}\{e^{iat} f(t)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (e^{iat} f(t)) e^{-ist} \, \mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i(s-a)t} \, \mathrm{d}t = F(s - a)$$

Property 2.3 (Time/Space Shift)**.** If $\mathcal{F}\{f(x)\} = F(s)$ and a is a real constant:

$$\mathcal{F}\{f(x-a)\} = e^{-isa}F(s)$$

Proof: Let $u = t - a$, so $t = u + a$ and $\mathrm{d}t = \mathrm{d}u$.

$$\begin{aligned}\mathcal{F}\{f(t-a)\} &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t-a)e^{-ist}\,\mathrm{d}t \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(u)e^{-is(u+a)}\,\mathrm{d}u \\ &= e^{-isa}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(u)e^{-isu}\,\mathrm{d}u\right) = e^{-isa}F(s)\end{aligned}$$

Property 2.4 (Scaling)**.** If $\mathcal{F}\{f(x)\} = F(s)$ and $a \neq 0$ is a real constant:

$$\mathcal{F}\{f(ax)\} = \frac{1}{|a|}F\left(\frac{s}{a}\right)$$

Proof: Let $u = at$, so $\mathrm{d}t = \mathrm{d}u/a$. If $a > 0$, the limits of integration remain $-\infty$ to ∞.

$$\mathcal{F}\{f(at)\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(at)e^{-ist}\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(u)e^{-is(u/a)}\frac{\mathrm{d}u}{a} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

If $a < 0$, the limits are reversed, which introduces a negative sign that is absorbed by a. The general result is $1/|a|$.

Property 2.5 (Symmetry Properties)**.** Let $\mathcal{F}\{f(x)\} = F(s)$.

1. **Time/Space Reversal:** $\mathcal{F}\{f(-x)\} = F(-s)$
2. **Complex Conjugation:** $\mathcal{F}\{f^*(x)\} = F^*(-s)$, where * denotes the complex conjugate.

Property 2.6 (Multiplication by Cosine and Sine)**.** Using Euler's formula, $e^{iax} = \cos(ax) + i\sin(ax)$, and linearity:

$$\begin{aligned}\mathcal{F}\{f(x)\cos(ax)\} &= \mathcal{F}\left\{f(x)\frac{e^{iax}+e^{-iax}}{2}\right\} = \frac{1}{2}[F(s-a)+F(s+a)] \\ \mathcal{F}\{f(x)\sin(ax)\} &= \mathcal{F}\left\{f(x)\frac{e^{iax}-e^{-iax}}{2i}\right\} = \frac{1}{2i}[F(s-a)-F(s+a)]\end{aligned}$$

Property 2.7 (Differentiation in Time/Space Domain)**.** If $f(x)$ and its derivatives vanish at $\pm\infty$:

$$\mathcal{F}\{f^{(n)}(x)\} = (is)^n F(s)$$

Property 2.8 (Differentiation in Frequency Domain)**.**

$$\mathcal{F}\{x^n f(x)\} = (i)^n \frac{d^n}{ds^n}F(s)$$

3 Convolution Theorem

Definition 3.1 (Convolution)**.** The convolution of two functions $f(x)$ and $g(x)$, denoted $(f * g)(x)$, is defined as:

$$(f*g)(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)g(x-t)\,\mathrm{d}t$$

Note that convolution is commutative: $f * g = g * f$.

Theorem 3.1 (Convolution Theorem)**.** The Fourier transform of a convolution is the product of their Fourier transforms.

$$\mathcal{F}\{(f * g)(x)\} = F(s)G(s)$$

Proof:

$$\begin{aligned}\mathcal{F}\{(f * g)(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) g(x-t)\, \mathrm{d}t \right] e^{-isx}\, \mathrm{d}x \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left[\int_{-\infty}^{\infty} g(x-t) e^{-isx}\, \mathrm{d}x \right] \mathrm{d}t \quad \text{(by Fubini's Theorem)}\end{aligned}$$

In the inner integral, let $u = x - t$, so $x = u + t$ and $\mathrm{d}x = \mathrm{d}u$.

$$\begin{aligned}\int_{-\infty}^{\infty} g(x-t) e^{-isx}\, \mathrm{d}x &= \int_{-\infty}^{\infty} g(u) e^{-is(u+t)}\, \mathrm{d}u = e^{-ist} \int_{-\infty}^{\infty} g(u) e^{-isu}\, \mathrm{d}u \\ &= e^{-ist} \sqrt{2\pi} G(s)\end{aligned}$$

Substituting this back:

$$\begin{aligned}\mathcal{F}\{(f * g)(x)\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left[e^{-ist} \sqrt{2\pi} G(s) \right] \mathrm{d}t \\ &= \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist}\, \mathrm{d}t \right) G(s) \\ &= F(s)G(s)\end{aligned}$$

4 Parseval's and Plancherel's Theorems

These theorems relate the energy of a signal in the time/space domain to its energy in the frequency domain.

Theorem 4.1 (Parseval's Identity)**.** If $\mathcal{F}\{f(x)\} = F(s)$ and $\mathcal{F}\{g(x)\} = G(s)$, then:

$$\int_{-\infty}^{\infty} f(x) g^*(x)\, \mathrm{d}x = \int_{-\infty}^{\infty} F(s) G^*(s)\, \mathrm{d}s$$

A special case is when $g(x) = f(x)$, which gives Plancherel's theorem.

Theorem 4.2 (Plancherel's Theorem)**.** The total energy of a function is equal to the total energy of its spectrum.

$$\int_{-\infty}^{\infty} |f(x)|^2\, \mathrm{d}x = \int_{-\infty}^{\infty} |F(s)|^2\, \mathrm{d}s$$

Proof Sketch: Start with $\int |f(x)|^2\, \mathrm{d}x = \int f(x) f^*(x)\, \mathrm{d}x$. Substitute the inverse Fourier transform for $f(x)$.

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) f^*(x)\, \mathrm{d}x &= \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{isx}\, \mathrm{d}s \right] f^*(x)\, \mathrm{d}x \\ &= \int_{-\infty}^{\infty} F(s) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f^*(x) e^{isx}\, \mathrm{d}x \right] \mathrm{d}s\end{aligned}$$

The inner integral is $\left(\frac{1}{\sqrt{2\pi}} \int f(x) e^{-isx}\, \mathrm{d}x \right)^* = F^*(s)$. Thus, the expression becomes $\int F(s) F^*(s)\, \mathrm{d}s = \int |F(s)|^2\, \mathrm{d}s$.

5 Important Fourier Transform Pairs (Examples)

Example 5.1 (Rectangular Pulse)**.** Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$.

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} 1 \cdot e^{-ist} \, \mathrm{d}t = \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-ist}}{-is} \right]_{-a}^{a}$$
$$= \frac{1}{\sqrt{2\pi}} \frac{e^{-isa} - e^{isa}}{-is} = \frac{1}{\sqrt{2\pi}} \frac{2\sin(as)}{s} = \sqrt{\frac{2}{\pi}} \frac{\sin(as)}{s}$$

This is often written using the unnormalized sinc function, $\operatorname{sinc}(z) = \sin(z)/z$, as $F(s) = \sqrt{\frac{2}{\pi}} a \operatorname{sinc}(as)$.

Deduction 1: Evaluate $\int_{-\infty}^{\infty} \frac{\sin t}{t} \, \mathrm{d}t$. Use the inverse transform for $f(x)$ at $x = 0$. We know $f(0) = 1$.

$$f(0) = 1 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^0 \, \mathrm{d}s = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi}} \frac{\sin(as)}{s} \, \mathrm{d}s = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(as)}{s} \, \mathrm{d}s$$

Thus, $\int_{-\infty}^{\infty} \frac{\sin(as)}{s} \, \mathrm{d}s = \pi$. Let $t = as$, $\mathrm{d}s = \mathrm{d}t/a$.

$$\int_{-\infty}^{\infty} \frac{\sin t}{t/a} \frac{\mathrm{d}t}{a} = \int_{-\infty}^{\infty} \frac{\sin t}{t} \, \mathrm{d}t = \pi$$

Since $\sin(t)/t$ is an even function, $\int_{0}^{\infty} \frac{\sin t}{t} \, \mathrm{d}t = \frac{\pi}{2}$.

Deduction 2: Evaluate $\int_{-\infty}^{\infty} \left(\frac{\sin t}{t}\right)^2 \mathrm{d}t$. Use Plancherel's theorem: $\int |f(x)|^2 \, \mathrm{d}x = \int |F(s)|^2 \, \mathrm{d}s$.

$$\int_{-a}^{a} |1|^2 \, \mathrm{d}x = 2a$$

$$\int_{-\infty}^{\infty} \left| \sqrt{\frac{2}{\pi}} \frac{\sin(as)}{s} \right|^2 \mathrm{d}s = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2(as)}{s^2} \, \mathrm{d}s$$

Equating them: $2a = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2(as)}{s^2} \, \mathrm{d}s \implies \int_{-\infty}^{\infty} \frac{\sin^2(as)}{s^2} \, \mathrm{d}s = \pi a$. Let $t = as$, then $\mathrm{d}s = \mathrm{d}t/a$.

$$\int_{-\infty}^{\infty} \frac{\sin^2 t}{(t/a)^2} \frac{\mathrm{d}t}{a} = a \int_{-\infty}^{\infty} \left(\frac{\sin t}{t}\right)^2 \mathrm{d}t = \pi a \implies \int_{-\infty}^{\infty} \left(\frac{\sin t}{t}\right)^2 \mathrm{d}t = \pi$$

Example 5.2 (Gaussian Function)**.** Find the Fourier transform of $f(x) = e^{-a^2 x^2}$ for $a > 0$.

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a^2 t^2} e^{-ist} \, \mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(a^2 t^2 + ist)} \, \mathrm{d}t$$

Complete the square in the exponent: $a^2 t^2 + ist = \left(at + \frac{is}{2a}\right)^2 - \left(\frac{is}{2a}\right)^2 = \left(at + \frac{is}{2a}\right)^2 + \frac{s^2}{4a^2}$.

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(at + \frac{is}{2a}\right)^2 - \frac{s^2}{4a^2}} \, \mathrm{d}t = \frac{e^{-s^2/4a^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(at + \frac{is}{2a}\right)^2} \, \mathrm{d}t$$

Let $u = at + \frac{is}{2a}$, so $\mathrm{d}t = \mathrm{d}u/a$. This is a contour integral, but it can be shown that the integral over the complex path is the same as over the real axis.

$$F(s) = \frac{e^{-s^2/4a^2}}{a\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2} \, \mathrm{d}u$$

Using the known Gaussian integral $\int_{-\infty}^{\infty} e^{-u^2}\,\mathrm{d}u = \sqrt{\pi}$:

$$F(s) = \frac{e^{-s^2/4a^2}}{a\sqrt{2\pi}}\sqrt{\pi} = \frac{1}{a\sqrt{2}}e^{-s^2/4a^2}$$

Self-Reciprocal Case: If we set $a^2 = 1/2$, so $a = 1/\sqrt{2}$, then $f(x) = e^{-x^2/2}$. The transform is $F(s) = \frac{1}{(1/\sqrt{2})\sqrt{2}}e^{-s^2/(4(1/2))} = e^{-s^2/2}$. Thus, $f(x) = e^{-x^2/2}$ is self-reciprocal under the Fourier transform.

Example 5.3 (Exponential Decay)**.** Find the Fourier transform of $f(x) = e^{-a|x|}$ for $a > 0$.

$$\begin{aligned}
F(s) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-a|t|}e^{-ist}\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}}\left[\int_{-\infty}^{0} e^{at}e^{-ist}\,\mathrm{d}t + \int_{0}^{\infty} e^{-at}e^{-ist}\,\mathrm{d}t\right] \\
&= \frac{1}{\sqrt{2\pi}}\left[\int_{-\infty}^{0} e^{(a-is)t}\,\mathrm{d}t + \int_{0}^{\infty} e^{-(a+is)t}\,\mathrm{d}t\right] \\
&= \frac{1}{\sqrt{2\pi}}\left(\left[\frac{e^{(a-is)t}}{a-is}\right]_{-\infty}^{0} + \left[\frac{e^{-(a+is)t}}{-(a+is)}\right]_{0}^{\infty}\right) \\
&= \frac{1}{\sqrt{2\pi}}\left(\frac{1}{a-is} - 0 + 0 - \frac{1}{-(a+is)}\right) = \frac{1}{\sqrt{2\pi}}\left(\frac{1}{a-is} + \frac{1}{a+is}\right) \\
&= \frac{1}{\sqrt{2\pi}}\frac{a+is+a-is}{a^2+s^2} = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}
\end{aligned}$$

Example 5.4 (FT of $xe^{-a|x|}$)**.** Using the differentiation in frequency property, $\mathcal{F}\{tf(t)\} = i\frac{d}{ds}F(s)$. Let $f(x) = e^{-a|x|}$, then $F(s) = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}$.

$$\begin{aligned}
\mathcal{F}\{xe^{-a|x|}\} &= i\frac{d}{ds}\left[\sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}\right] = i\sqrt{\frac{2}{\pi}}a\frac{d}{ds}(a^2+s^2)^{-1} \\
&= i\sqrt{\frac{2}{\pi}}a(-1)(a^2+s^2)^{-2}(2s) = -i\sqrt{\frac{2}{\pi}}\frac{2as}{(a^2+s^2)^2}
\end{aligned}$$

6 Fourier Cosine and Sine Transforms

For functions defined only on $[0,\infty)$, or for even/odd functions, the Fourier cosine and sine transforms are useful. They arise from considering the Fourier transform of even and odd extensions of a function $f(x)$.

6.1 Definitions

Let $f(x)$ be a function defined for $x \geq 0$.

Definition 6.1 (Fourier Cosine Transform)**.**

$$F_c(s) = \mathcal{F}_c\{f(x)\} = \sqrt{\frac{2}{\pi}}\int_0^{\infty} f(t)\cos(st)\,\mathrm{d}t$$

Its inverse is:

$$f(x) = \mathcal{F}_c^{-1}\{F_c(s)\} = \sqrt{\frac{2}{\pi}}\int_0^{\infty} F_c(s)\cos(sx)\,\mathrm{d}s$$

Definition 6.2 (Fourier Sine Transform)**.**

$$F_s(s) = \mathcal{F}_s\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \sin(st)\, \mathrm{d}t$$

Its inverse is:

$$f(x) = \mathcal{F}_s^{-1}\{F_s(s)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty F_s(s) \sin(sx)\, \mathrm{d}s$$

Note 6.1. A function is self-reciprocal under the cosine (sine) transform if $\mathcal{F}_c\{f\} = f$ ($\mathcal{F}_s\{f\} = f$).

Note 6.2 (Reciprocity)**.** The transform relations are symmetric. If $\mathcal{F}_c\{f(x)\} = F_c(s)$, then $\mathcal{F}_c\{F_c(x)\} = f(s)$. The same holds for the sine transform. This is a powerful property for finding new transform pairs.

6.2 Properties of Cosine and Sine Transforms

- **Linearity:** Both transforms are linear.
- **Differentiation:** If $f(x) \to 0$ as $x \to \infty$:

$$\mathcal{F}_c\{f'(x)\} = s\mathcal{F}_s\{f(x)\} - \sqrt{\frac{2}{\pi}} f(0)$$
$$\mathcal{F}_s\{f'(x)\} = -s\mathcal{F}_c\{f(x)\}$$

- **Multiplication by x:**

$$\mathcal{F}_c\{xf(x)\} = \frac{d}{ds}\mathcal{F}_s\{f(x)\}$$
$$\mathcal{F}_s\{xf(x)\} = -\frac{d}{ds}\mathcal{F}_c\{f(x)\}$$

- **Parseval's Identities:**

$$\int_0^\infty f(x)g(x)\, \mathrm{d}x = \int_0^\infty F_c(s)G_c(s)\, \mathrm{d}s$$
$$\int_0^\infty f(x)g(x)\, \mathrm{d}x = \int_0^\infty F_s(s)G_s(s)\, \mathrm{d}s$$
$$\implies \int_0^\infty |f(x)|^2\, \mathrm{d}x = \int_0^\infty |F_c(s)|^2\, \mathrm{d}s = \int_0^\infty |F_s(s)|^2\, \mathrm{d}s$$

6.3 Examples of Cosine and Sine Transforms

Example 6.1 (Transforms of e^{-ax})**.** For $a > 0$, we have the standard integral results:

$$\int_0^\infty e^{-at}\cos(st)\, \mathrm{d}t = \frac{a}{a^2+s^2} \quad \text{and} \quad \int_0^\infty e^{-at}\sin(st)\, \mathrm{d}t = \frac{s}{a^2+s^2}$$

Therefore:

$$\mathcal{F}_c\{e^{-ax}\} = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}$$
$$\mathcal{F}_s\{e^{-ax}\} = \sqrt{\frac{2}{\pi}}\frac{s}{a^2+s^2}$$

Example 6.2 (Transforms of xe^{-ax})**.** Using the 'multiplication by x' properties:

$$\begin{aligned}\mathcal{F}_c\{xe^{-ax}\} &= \frac{d}{ds}\mathcal{F}_s\{e^{-ax}\} = \frac{d}{ds}\left(\sqrt{\frac{2}{\pi}}\frac{s}{a^2+s^2}\right) \\ &= \sqrt{\frac{2}{\pi}}\frac{(a^2+s^2)(1)-s(2s)}{(a^2+s^2)^2} = \sqrt{\frac{2}{\pi}}\frac{a^2-s^2}{(a^2+s^2)^2}\end{aligned}$$

$$\begin{aligned}\mathcal{F}_s\{xe^{-ax}\} &= -\frac{d}{ds}\mathcal{F}_c\{e^{-ax}\} = -\frac{d}{ds}\left(\sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}\right) \\ &= -\sqrt{\frac{2}{\pi}}a(-1)(a^2+s^2)^{-2}(2s) = \sqrt{\frac{2}{\pi}}\frac{2as}{(a^2+s^2)^2}\end{aligned}$$

Example 6.3 (Transform of $1/(1+x^2)$)**.** We can find $\mathcal{F}_c\{1/(1+x^2)\}$ using reciprocity. From the transform of e^{-ax}, let $a = 1$:

$$\mathcal{F}_c\{e^{-x}\} = \sqrt{\frac{2}{\pi}}\frac{1}{1+s^2}$$

Let $f(x) = e^{-x}$ and $F_c(s) = \sqrt{\frac{2}{\pi}}\frac{1}{1+s^2}$. The reciprocity theorem states $\mathcal{F}_c\{F_c(x)\} = f(s)$.

$$\mathcal{F}_c\left\{\sqrt{\frac{2}{\pi}}\frac{1}{1+x^2}\right\} = e^{-s}$$

By linearity:

$$\mathcal{F}_c\left\{\frac{1}{1+x^2}\right\} = \sqrt{\frac{\pi}{2}}e^{-s}$$

7 Exercises

Exercise 7.1. Obtain the Fourier transform of $f(x) = \begin{cases}1-x^2, & |x| \le 1 \\ 0, & |x| > 1\end{cases}$. Hence evaluate $\int_0^\infty \frac{x\cos x - \sin x}{x^3}\cos(x/2)\,\mathrm{d}x$.

Exercise 7.2. Find the Fourier transform of $f(x) = \begin{cases}|x|, & |x| \le a \\ 0, & |x| > a\end{cases}$ where $a > 0$.

Exercise 7.3. Find the Fourier transform of $f(x) = \begin{cases}a^2-x^2, & |x| \le a \\ 0, & |x| > a\end{cases}$ where $a > 0$.